

# Process or Transmit? An Energy- and Thermal-Aware Onboard Decision Engine for LEO Satellite Data Handling

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**Abstract**—Small Earth-observation satellites capture multi-spectral and Synthetic Aperture Radar data under intermittent contact and constrained onboard resources. This paper studies the IASTAM 6.0 “Process or Transmit?” problem using a seeded, one-second-step LEO simulator and its implemented threshold heuristic. Across five four-orbit episodes (seeds 42–46), the heuristic achieved a mean delivered-utility ratio of 0.0283 (sample standard deviation 0.0024; 95% t interval 0.0252–0.0313) and completed 1.53% of generated payloads. Mean modeled energy draw was 291.41 kJ, mean battery state of charge stayed above the 30% floor, and no safe-mode entry occurred. In a separate two-orbit trace (seed 42), simulated chip temperature reached 52.48°C; the simulator throttles compute at 50°C but does not enforce a hard temperature cap. Learned-policy, Lyapunov-scheduler, oracle, and hardware-calibration studies remain future work.

**Index Terms**—orbital edge computing, onboard task scheduling, energy- and thermal-aware computing, Lyapunov optimization, reinforcement learning, Low Earth Orbit, fault tolerance.

## I. INTRODUCTION

The 2026 edition of the IEEE IAS Tunisia Annual Meeting (IASTAM 6.0) Technical Challenge, organized in collaboration with the Tunisian Space Association (TUNSA), addresses an operational bottleneck in Earth-observation data handling. Orbital edge computing has drawn attention as a way to move selected processing closer to sensors [1], [8]. Google’s Project Suncatcher is a proposed space-based AI infrastructure concept, not an operational orbital testbed [8].

Within this domain, Problem 1 (“Process or Transmit?”) targets an acute bottleneck in Earth Observation (EO): small satellites continuously capture high-resolution optical and Synthetic Aperture Radar (SAR) payloads (generating multi-megabyte payloads every 15 s), yet operate under three binding physical constraints:

- 1) **Battery Floor:** The simulator models a 100 Wh battery and clamps its state at a 30% minimum depth-of-discharge floor.
- 2) **Thermal Constraint:** Vacuum requires radiative heat rejection. The current simulator uses a first-order thermal

proxy and pauses compute at 50°C; it does not enforce a hard cap, and the measured trace reaches 52.48°C.

- 3) **Severe Downlink Bandwidth Choke:** Line-of-sight contact with ground stations is limited to short windows (120–180 s per 90-minute orbit, about 2.8% mean contact duty cycle) with an X-band rate of 20.0 Mbps (2.5 MB/s).

For every captured payload, the onboard system must choose among four actions: `PROCESS_NOW` (compression vs. inference), `STORE_QUEUE` (time-shifted execution in Mass Memory Unit, MMU), `TRANSMIT_RAW` (downlink during line-of-sight passes), or `DISCARD_DROP` (pruning low-utility or cloud-occluded data). The optimization objective is to maximize cumulative scientific and operational utility, which decays exponentially over latency.

### Contributions:

- We characterize the repository’s seeded one-second simulator and implemented multi-threshold heuristic.
- We report five four-orbit heuristic episodes with decision quality, task completion, energy, latency, MMU/RAM utilization, SEU counts, and safe-mode events.
- We identify an observed temperature excursion above the code’s 50°C throttling threshold, a limitation that motivates thermal-model and safety-guard validation.

## II. RELATED WORK

### A. Orbital and Edge Computing Systems

Orbital edge computing surveys describe the systems-level case for in-space processing [1]. Trabant is a recent serverless architecture proposal [2]; the Tiansuan constellation provides a cloud-native satellite case study [3]. Broader treatments of Space AI [4] and Earth-observation image processing [5] provide context on onboard classification and inference. FPGA accelerator surveys [6] and work on when to compute in space [7] inform the modeled trade-offs; the simulator’s numerical parameters remain assumptions rather than transferred empirical measurements.

### B. Control and Scheduling Paradigms

Resource-constrained orbital scheduling is addressed via two dominant paradigms:

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- 1) **Lyapunov Optimization & Drift-Plus-Penalty:** Prior work [10] motivates online scheduling; implementing and evaluating an orbital scheduler remains planned work.
- 2) **Constrained Reinforcement Learning (CMDP):** The repository includes a Gymnasium environment; policy training and safety evaluation are not reported here.

### C. Added Value of This Work

Prior work motivates orbital edge processing and constrained scheduling. This interim study contributes a reproducible characterization of the current heuristic and identifies model and benchmarking gaps.

## III. METHODOLOGY

### A. Physical Models & Orbital Mechanics

The satellite orbital environment and hardware state at time  $t$  are formalized in Table I.

TABLE I  
PRIMARY SYSTEM AND STATE VARIABLES

| Symbol                                   | Physical Definition            | Nominal Value / Range                                         |
|------------------------------------------|--------------------------------|---------------------------------------------------------------|
| $\tau_{\text{orbit}}, \tau_{\text{sun}}$ | Orbit period / Sunlit duration | 5400 s / 3300 s (90 / 55 min)                                 |
| $P_{\text{harvest}}(t)$                  | Solar power harvested          | 0.0 W (ecl) to 30.0 W peak                                    |
| $\text{SoC}(t)$                          | Battery State-of-Charge        | Floor $\geq 0.30$ , Cap: 100.0 Wh                             |
| $T_{\text{chip}}(t)$                     | Processor temperature          | Ambient $0^\circ\text{C}$ , Ceiling $\leq 50.0^\circ\text{C}$ |
| $G(t)$                                   | Ground station contact flag    | $\{0, 1\}$ , 120–180 s pass (about 2.8% mean duty)            |
| $B_{\text{downlink}}$                    | RF communication speed         | 20.0 Mbps = 2.5 MB/s (15.0 W)                                 |
| $P_{\text{seu}}$                         | SEU bit-flip rate              | $10^{-5} \text{ s}^{-1}$ cosmic radiation                     |
| $M_{\text{mmu}}, M_{\text{ram}}$         | Mass Memory / RAM              | 32 000 MB / 4 000 MB                                          |

1) **Power and Battery Dynamics:** The simulator uses a 55-minute sunlight interval followed by a 35-minute eclipse. In sunlight, harvested power follows a sine-shaped profile from a 22.5 W base to a 30 W peak, with 30 s cosine tapers at sunlight boundaries; eclipse harvesting is zero.

Battery energy is updated by

$$E_{\text{batt}}(t+1) = \min(E_{\text{cap}}, E_{\text{batt}}(t) + [P_{\text{harvest}}(t) - P_{\text{draw}}(t)]\Delta t), \quad (1)$$

where  $P_{\text{draw}} = P_{\text{base}} + P_{\text{compute}} + P_{\text{comm}}$ ,  $P_{\text{base}} = 5 \text{ W}$ , and radio draw is 15 W during downlink. The implementation has no charge/discharge efficiency; it clamps battery energy to the 30% floor on brownout and exits safe mode above 1.05 times that floor.

2) **Lumped Thermal Proxy:** The simulator uses a first-order thermal proxy rather than a Stefan–Boltzmann radiation model:

$$T_{\text{chip}}(t+1) = T_{\text{chip}}(t) + 0.15P_{\text{draw}}(t) - 0.10(T_{\text{chip}}(t) - T_{\text{amb}}). \quad (2)$$

This one-second update uses watts of total bus, compute, and communication draw, with  $T_{\text{amb}} = 0^\circ\text{C}$ . Compute pauses at or above  $50^\circ\text{C}$ , but this threshold is not a hard cap; the seed-42 two-orbit trajectory peaks at  $52.48^\circ\text{C}$ .

### B. Workload, Queueing, and Utility Formulation

Payloads  $k$  arrive periodically every  $\Delta t_{\text{gen}} = 15 \text{ s}$  across dual modalities:

- **Optical Multispectral (70%):** Size  $S_k \in [5, 15] \text{ MB}$ , initial utility  $U_{0,k} \in [10, 100]$ .
- **Synthetic Aperture Radar (30%):** Size  $S_k \in [100, 300] \text{ MB}$ , initial utility  $U_{0,k} \in [10, 100]$ .

Payload utility decays exponentially over latency ( $t - t_{\text{capture}}$ ):

$$U_k(t) = U_{0,k} \cdot \exp(-\alpha_k(t - t_{\text{capture}})) \quad (3)$$

with decay parameter  $\alpha_k \in [10^{-4}, 10^{-3}] \text{ s}^{-1}$ .

Compute actions transform payload properties:

- **Compression:** Duration 2 s,  $P_{\text{comp}} = 10 \text{ W}$ , size factor  $C_{\text{ratio}} = 0.30$ , utility retention  $\rho_{\text{val}} = 0.90$ .
- **Inference:** Duration 5 s (with 2 s wake penalty),  $P_{\text{inf}} = 20 \text{ W}$  (30 W wake), size factor  $C_{\text{ratio}} = 0.001$ , utility retention  $\rho_{\text{val}} = 0.50$ .

### C. Decision Engine & Solution Schedulers

For each payload  $k$ , the agent selects an action  $a_k \in \mathcal{A}$ , where:

$$\mathcal{A} = \{\text{PROCESS\_COMPRESS}, \text{PROCESS\_INFER}, \text{DOWNLINK\_RAW}, \text{STORE\_QUEUE}, \text{DISCARD\_DROP}\} \quad (4)$$

- 1) **Baseline Heuristic Policy:** During ground contact it requests one highest-ranked payload per tick, respecting the shared link budget. When  $\text{SoC} > 0.50$ ,  $T_{\text{chip}} < 50^\circ\text{C}$ , and queue depth  $< 2$ , it executes inference (for payloads  $< 50 \text{ MB}$ ) or compression. Stale payloads ( $U_k(t) < 5.0$ ) are evicted when MMU usage  $> 80\%$ .
- 2) **Constrained MDP (Gymnasium):** State  $s_t \in \mathbb{R}^{13+5K}$  captures normalized telemetry and top- $K$  payload meta-data. Invalid action masks zero out physically infeasible actions.
- 3) **Planned Lyapunov Scheduler:** Drift-plus-penalty scheduling is a Phase 3 target and is not implemented in the current repository.

## IV. EXPERIMENTS AND RESULTS

### A. Experimental Setup and Results

The implemented heuristic was evaluated for five four-orbit episodes (21,600 one-second steps per episode; seeds 42–46), generating 1,440 payloads in each episode. Table II reports means, sample standard deviations, and 95% Student- $t$  confidence intervals ( $n = 5$ ). The evaluator’s decision-quality denominator is the snapshot sum of generated payload base values. Memory utilization is time-averaged occupancy; energy includes base, compute, and communication draw; latency averages completed deliveries only.

The SEU confidence interval is bounded below by zero; five episodes are too few to estimate rare-event rates reliably. Across a separate two-orbit seed-42 trace, SoC ranged from

TABLE II  
IMPLEMENTED HEURISTIC ACROSS FIVE FOUR-ORBIT EPISODES

| Metric                     | Mean    | Sample SD | 95% CI             |
|----------------------------|---------|-----------|--------------------|
| Decision quality           | 0.0283  | 0.0024    | [0.0252, 0.0313]   |
| Completed-task rate        | 0.0153  | 0.0010    | [0.0141, 0.0165]   |
| Cumulative energy (kJ)     | 291.414 | 1.850     | [289.117, 293.711] |
| Mean latency (s)           | 654.00  | 136.05    | [485.10, 822.89]   |
| MMU utilization (%)        | 37.497  | 2.094     | [34.897, 40.097]   |
| RAM utilization (%)        | 0.172   | 0.002     | [0.169, 0.175]     |
| SEU events / episode       | 0.400   | 0.548     | [0.000, 1.080]     |
| Safe-mode events / episode | 0.000   | 0.000     | [0.000, 0.000]     |

92.37% to 100%, while temperature ranged from 0.75°C to 52.48°C. The 50°C threshold throttles processing but does not bound the simulated temperature.

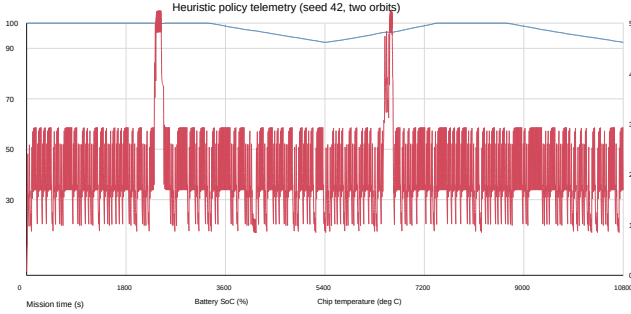


Fig. 1. Heuristic trajectory over two orbits (seed 42). Temperature exceeds the 50°C throttle threshold.

This is a single-policy characterization. The current code-base does not implement the claimed transmit-all comparison, Lyapunov scheduler, or MILP oracle; comparative policy results are future work.

### B. Phase 3 Validation Plan

Phase 3 development will: (1) train a MaskablePPO agent in *simulation/rl\_env.py*; (2) integrate the Lyapunov safety filter to bound RL proposals; (3) compute the MILP oracle ceiling; and (4) stress-test under solar flare SEU bursts ( $p_{\text{seu}} = 10^{-3}$ ) and degraded battery packs (60 Wh).

## V. CONCLUSION

We report a reproducible five-seed, four-orbit characterization of the implemented heuristic. The measured decision-quality ratio is 0.0283 on the current simulator workload. The two-orbit trace also exposes a thermal-model limitation: simulated temperature can exceed the 50°C throttling threshold. Future work includes comparative baselines, a calibrated thermal model with a hard safety guard, and implementation of Lyapunov and oracle schedulers.

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